# Appendix A: Research Protocol

## Study 1: Validating SpeechLP's Automated Speech Scoring Against SLP Ratings, Acoustic Measures, and Parent Judgments

Protocol draft, version 0.1 · November 2026 · Prepared by SpeechLP for review by prospective research partners. Items marked in yellow are proposals to be finalized with the partner.

| | |
|---|---|
| **Design** | Prospective, cross-sectional measurement-validation study comparing the app's accept/reject decisions with a reference standard built from blinded SLP ratings. |
| **Participants** | 40 English-speaking children aged 3–12, spanning a range of speech motor skill. |
| **Primary outcome** | Agreement between app decisions and SLP consensus: accuracy and unweighted average recall (UAR). |
| **Lead** | University partner (design, ethics approval, data collection, analysis); SpeechLP provides the app, engineering support, and recruitment through connected SLPs. |
| **Duration** | About [9] months from ethics approval to manuscript submission. |

### A.1 Background and rationale

SpeechLP uses automated speech recognition to decide, for each practice card, whether a child's production of the target word is accepted. In a retrospective study of 101 returning profiles, mean accepted-card accuracy rose from 69.4% to 74.9% across two months (Silver et al., 2026), but the authors could not interpret this as clinical improvement because the acceptance model had not been compared with SLP judgment. Every later study of SpeechLP's effects depends on knowing how far its scores can be trusted, so validation comes first.

A Swedish study of 28 children aged 4–10 found that a speech recognition model rated single words within about 5 percentage points of an expert's own re-ratings on accuracy, although the expert was better at distinguishing the middle levels of the scale (Getman et al., 2022). That design has not been tested in English or in home practice. Because expert listening has its own limits (Meyer & Munson, 2021; Allison et al., 2025), this study uses several SLP raters, reports their agreement with each other, and adds instrumental (acoustic) measures (Namasivayam et al., 2025).

SpeechLP is working to fine-tune its speech recognition model. By showing where agreement with SLPs is lowest, for example by sound, age, or error type, the results will guide that work. Any use of study recordings for model development would require explicit consent and would be covered by the data use agreement.

## A.2 Research questions

1. **Primary.** How closely do SpeechLP's accept/reject decisions agree with consensus correct/incorrect ratings from blinded SLPs?
2. How does that agreement compare with the agreement between individual SLPs and the consensus of the other raters?
3. Does agreement differ by target sound, word position, child age, speech motor skill, and error type (substitution versus distortion)?
4. How do parents' in-the-moment correct/incorrect judgments during practice compare with the SLP consensus and with the app?
5. For sounds with established acoustic correlates, how do app decisions relate to acoustic measures?

## A.3 Participants

**Inclusion.** Children aged 3–12 years who speak English as a primary language, with parent or guardian consent and child assent appropriate to age. The sample will deliberately include children with identified speech sound disorders (including residual errors such as /ɹ/ and /s/ distortions) and children with typically developing speech, so that the app is tested on both correct and incorrect productions.

**Exclusion.** Uncorrected hearing loss, structural or neurological conditions that prevent participation in a picture-naming task, or inability to complete practice on a tablet or computer. [Partner to confirm.]

**Recruitment.** Through SLPs connected to SpeechLP via SpeechTherapist.app and through the partner's clinic or school networks. Recruitment will be stratified by age band (3–5, 6–8, 9–12) and by speech status.

**Sample size.** 40 children, each contributing about 120 scored productions (roughly 4,800 in total). This exceeds the 28 children in Getman et al. (2022), extends the age range to include older children with residual errors, whom SpeechLP already serves, and allows agreement to be estimated by age band and for frequently targeted sounds. A precision calculation for the primary UAR estimate, accounting for clustering of productions within children, will confirm that this sample is adequate before recruitment begins.

## A.4 Procedures

1. **Baseline assessment.** A partner SLP administers a brief standardized single-word articulation measure, chosen by the research partner, from which percentage of consonants correct (PCC; Shriberg et al., 1997) is calculated, plus stimulability testing for error sounds. Diagnosis and severity are recorded where available.
2. **Practice sessions.** Each child completes two sessions (about 60 productions each) of standard SpeechLP practice on a study iPad or a computer (web version), at home or in clinic, using a fixed word list that samples all 24 consonants across word positions, weighted toward the child's error sounds.
3. **Recording.** The partner records each session with a research-grade microphone under its ethics approval (SpeechLP stores no audio on its servers). The study build timestamps each decision and each play or replay of the spoken model, so recordings align with app decisions word by word and each production can be classed as imitated or spontaneous.
4. **Parent judgment.** A parent sitting with the child marks each production correct or incorrect at the time of practice, without seeing the app's decision first, on a numbered study form, writing each word shown and marking it correct or incorrect; since games vary word order, entries are matched to the app's timestamped log.
5. **Blinded SLP rating.** At least three certified SLPs independently rate every production from the recordings, blinded to the app decision, the parent judgment, and each other. Each production receives a binary correct/incorrect rating and a rating on the five-level scale used by Getman et al. (2022), from unidentifiable to correct, including subphonemic errors. Raters complete a calibration set before rating, and about 20% of productions are re-presented to estimate intra-rater reliability.
6. **Acoustic analysis.** For sounds with established acoustic correlates (for example, formant measures for /ɹ/ and spectral measures for /s/ and /ʃ/), trained analysts blinded to all ratings extract acoustic measures. [Partner to specify sounds and measures.]

## A.5 Measures

| | |
|---|---|
| **App decision** | Accepted or not accepted for each production, with the sounds recognized, app version, and platform (iOS app or web). |
| **Model exposure** | Whether the spoken model was played or replayed before each production, and how many times. |
| **Reference standard** | Majority vote of SLP raters (binary) and median across raters (five-level); uncertainty in the reference is reported. |
| **Child characteristics** | Age, PCC, stimulability, diagnosis and severity where available, and rater-coded error type. |

### A.6 Analysis plan

The analysis plan and the hypotheses below will be preregistered (for example, on the Open Science Framework) before any ratings are compared with app decisions. This addresses a limitation of the retrospective study, whose comparisons were defined after the data were inspected.

- **Primary analysis.** Accuracy and unweighted average recall (the mean of sensitivity and specificity, which is robust to the imbalance between correct and incorrect productions) of app decisions against the SLP consensus, with 95% confidence intervals from a bootstrap that resamples whole children to respect clustering. Sensitivity, specificity, and Cohen's kappa are reported alongside. Where the app produces a graded score, mean absolute error against the five-level rating is also reported, matching the metrics of Getman et al. (2022).
- **Human benchmark (RQ2).** Each SLP's agreement with the consensus of the other raters is calculated with the same metrics, giving a range of human performance against which the app is compared. Inter-rater reliability is reported as Fleiss' kappa (binary) and an intraclass correlation (five-level).
- **Moderators (RQ3).** Mixed-effects logistic regression of app–consensus agreement, with random intercepts for child and word, and fixed effects for sound, word position, age band, PCC, error type, and model exposure (imitated versus spontaneous production), and platform (iOS app or web). Agreement is also reported separately for imitated and spontaneous productions. Sound-level results with fewer than [20] productions are reported descriptively only.
- **Parents (RQ4).** Parent judgments are compared with the SLP consensus and with the app using the primary metrics.
- **Acoustics (RQ5).** Acoustic measures are compared between accepted and rejected productions, and with the five-level SLP ratings, within each analyzed sound.
- **Decision rule.** Before data collection, the partner and SpeechLP will agree on the agreement level at which app scores are considered adequate for use as an outcome measure, for example UAR within [X] percentage points of the median individual SLP.

### A.7 Ethics, privacy, and data management

- Ethics approval through the partner's institutional review board, with written parental consent and child assent.
- Study recordings are collected and held by the partner under its approval. SpeechLP does not receive audio unless participants separately and explicitly consent to share recordings for model development, as covered by the data use agreement.
- App decision logs are shared de-identified under a data use agreement. Account and child identifiers never leave SpeechLP, and small groups are suppressed in any reporting.

### A.8 Roles and independence

The partner leads study design, ethics approval, data collection, and analysis, and may publish regardless of results. SpeechLP provides app licenses, a study build that logs decisions and model replays with timestamps, engineering time for data exports, and recruitment through connected SLPs. SpeechLP reviews manuscripts within 30 days only to protect participant privacy and to correct factual errors about the product. SpeechLP's cofounders have a financial interest in the app, which will be disclosed in all outputs; SpeechLP clinicians may be co-authors where their contributions warrant it.

### A.9 Proposed timeline

| | |
|---|---|
| **Months 1–2** | Finalize protocol, obtain ethics approval, preregister, prepare study build and word list. |
| **Months 3–5** | Recruitment, baseline assessment, and practice sessions. |
| **Months 5–7** | Blinded SLP rating, reliability checks, and acoustic analysis. |
| **Months 8–9** | Preregistered analysis; manuscript and conference submission (for example, a co-authored ASHA Convention submission). |

### A.10 Study 2 (summary): Home practice dose using existing data

**Questions.** Do the amount, spacing, and accuracy of home practice predict gains, and how closely do families complete SLP-assigned practice? **Data.** De-identified practice records (cards, sessions, practice days, timestamps, target sounds, and SLP assignments through SpeechTherapist.app) shared under a data use agreement. **Analysis.** Preregistered mixed-effects growth models of accuracy over time, with practice exposure and spacing as predictors and starting accuracy adjusted for to limit regression to the mean; completion of assigned practice is described as the proportion of assigned sessions completed. **Interpretation.** Results are read in light of Study 1's validation and show associations, not causes.